

TALI BAR-KOHANY, Ph.D.

Curriculum vitae

2026

Personal Details

Name: Tali Bar-Kohany
Date & Place of birth: July 21, 1973, Beer-Sheva, ISRAEL
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Education

Ben-Gurion University of the Negev, Department of Mechanical Engineering

Ph.D.	2000 - 2004	A Study of the Mechanism of Sprays Formation by Bi-Component Liquid Flashing
M.Sc.	1998 - 2000	Strategic Analysis of Variable Valve Timing in SI Engines
B.Sc. <i>with honors</i>	1994 – 1998	Thermodynamical Analysis of a Practical Otto Cycle with Entropy and Availability
Online	09-10/ 7 /2020	LIDESP VIII 8 th International Advanced Course on Liquid Interfaces, Drops and Sprays.
VKI, Belgium	23-27/ 1 /2006	Introduction to CFD, Von-Karman Institute for Fluid Dynamics
CISM, Italy	10-14/ 7 /2000	Interfacial Phenomena, The Marangoni Effect, Instability, Waves and Convective Flows, International Center for Mechanical Science

Awards& Fellowship

2022	2 nd prize for presented poster at the 35 th Annual Symposium of the Israeli Section of the Combustion Institute
2016	1 st prize for presented poster at the 30 th Annual Symposium of the Israeli Section of the Combustion Institute
2010	Honorary Fellow, Australian Institute of High Energetic Materials
2004 - 2010	Katzir Scholarships Program – Ministry of Defense

2002	1 st prize for presented poster at the 18 th Annual Symposium of the Israeli Section of the Combustion Institute.
2001	Major Ehud Ben-Amitai Scholarships Fund for Aeronautic research
2000	WOLF award for Outstanding Master achievements
1998	Head of Department award for excellence

WORK EXPERIENCE

3/2023 – present	<i>Senior Researcher, A</i> (equivalent to Associate professor) Nuclear Research Center of the Negev R&D Engineering Department
8/2017 – 8/2018	<i>Visiting faculty & External Lecturer</i> University of Maryland, College Park Department of Mechanical Engineering
10/2015 – present	<i>Associate Professor (2026)</i> Tel-Aviv University School of Mechanical Engineering
8/2013 – 8/2014	<i>Visiting faculty</i> University of California, Irvine Department of Mechanical Engineering
7/2004 – 3/2023	<i>Senior Researcher</i> Nuclear Research Center of the Negev 2004 -2007: Physics Department 2007 – 2023: R&D Engineering Department Senior researcher: Two-phase & reactive flow, Phase change, Fluid Dynamics, Vacuum.
2008- 2017 2019	<i>Lecturer, External</i> Ben-Gurion University of the Negev First Course of Thermodynamics - Mechanical, Energy & Biotechnology Engineering. Principles in Combustion – Safety and Management Engineering.
1998 – 2004	<i>Teaching Assistant</i> Department of Mechanical Engineering, BGU Frontal and Lab instruction in the courses of Thermodynamics, Heat Transfer and Fluid Mechanics.

COURSES TAUGHT

2021 – 2022	<i>Lecturer and course development</i> School of Mechanical Engineering, TAU <u>Advanced Thermodynamics</u>
2021 – 2022	<i>Lecturer</i> School of Mechanical Engineering, TAU Second course of <u>Fluid Mechanics</u>
2020	<i>Lecturer</i> Department of BioMedical Engineering, TAU First course of <u>Thermodynamics</u>
2019 2008 – 2010	<i>Lecturer</i> Department of Mechanical Engineering, BGU First course of <u>Thermodynamics</u>
2017	<i>Lecturer</i> Department of Mechanical Engineering, UMD <u>Fluid Mechanics</u>
2015– 2017	<i>Lecturer</i> Energy Engineering Unit, BGU Introduction to energy engineering – <u>Thermodynamics</u>
2014	<i>Lecturer</i> Department of Biotechnology Engineering, BGU First course of <u>Thermodynamics</u>
2012– 2017	<i>Lecturer</i> Management and Safety Engineering Unit for Graduate studies, BGU, <u>Principles of Combustion processes</u>
2012	<i>Lecturer</i> Department of Mechanical Engineering, SCE First course of <u>Heat transfer</u>
2008 – 2010 2019	<i>Lecturer</i> Department of Mechanical Engineering, BGU First course of <u>Thermodynamics</u>
2004 – 2008	<i>Guest Lecturer</i> Department of Materials Engineering, BGU <u>Fundamentals of Leak Testing, for the course of Non-Destructive Evaluation</u>
2002 – 2004	<i>Lecturer, Developed syllabus and overall course structure</i> BGU School of Continued Education <u>Physical Fundamentals of Mechanical Engineering</u>

STUDENTS

2025-present	Tomer Lozano , M.Sc. in Mechanical Engineering TAU <i>An Experimental Study on the Effects of Size and Heating Rate on Critical Heat Flux</i>
2023-present	Omry Magen , Ph.D. in Mechanical Engineering TAU <i>Rapid phase-change of hydrogen and hydrogen carrier fuels (HFCs) - RaPhys</i>
2021-2025	Doron Biton , M.Sc. in Mechanical Engineering BGU, <i>Experimental study of the critical heat flux temperature due to rapid isobaric heating.</i>
2020-2023	Yogev Elfasi , M.Sc. Project in Mechanical Engineering TAU <i>Effect of wall-liquid intermolecular forces on nucleation in a rapid isobaric heating</i>
2019-2022	Amiav Lankry , M.Sc. in Mechanical Engineering BGU, <i>Nucleation inception temperature in boiling due to rapid heating under various thermodynamic conditions.</i>
2016-2020	Amir Shalel , M.Sc. in Safety and Management Engineering BGU, <i>Optimization of extinguishing powder dispersal.</i>
2016-2019	Lior Nahon , M.Sc. in Energy Engineering BGU <i>Countercurrent limited critical heat flux with heat transfer to the wall.</i>
2016-2019	Yarden Amsalem , M.Sc. in Mechanical Engineering TAU <i>Cubic Equations of State and nucleation inception in boiling due to rapid heating.</i>
2016-2018	Yaniv Levin , M.Sc. Project in Mechanical Engineering TAU <i>Modelling and simulating Ozonation of Lignocellulosic Biomass.</i>
2014-2016	Tal Eluk , M.Sc. in Mechanical Engineering BGU <i>Two-phase reactive flow; CFD-DPM model of combustion process in a co-current gas-solid reactor.</i>
2011 – 2015	Naor Hayun , M.Sc. in Mechanical Engineering BGU <i>Temperature profile control as a tool to improve chemical reaction in gas-particle downer.</i>
2013 – 2014	Idan Cohen , M.Sc. in Mechanical Engineering, BGU <i>Study of the thermal behavior of a thermo-luminescence detector (TLD) due to gas heating.</i>
2010 – 2012	Kobi Dahan , M.Sc. in Mechanical Engineering, BGU <i>Gas leakage and flammability into enclosures.</i>
2009 – 2011	Ravit Rozenfeld , M.Sc. in Mechanical Engineering, BGU <i>Study of the thermal behavior of a thermo-luminescence detector (TLD) due to gas heating.</i>

Scientific Publications

Journal Papers

<https://scholar.google.com/citations?user=QEXmTrTRGVQC&hl=en>

1. Magen, O., Kozak, Y., Di Lucchio, L., Marengo, M. and **Bar-Kohany, T.**, (2025), [Predicting nucleation pressure under rapid depressurization: Bridging positive and negative pressure regions](#). *International Journal of Heat and Mass Transfer*, 251, 127309.
2. Chernov, A. A., Antonov, D. V., Pavlenko, A. N., **Bar-Kohany, T.**, Strizhak, P. A., and Sazhin, S. S., (2025) "[Bubble growth in composite water/fuel droplets: effect on timing of their puffing/micro-explosion](#)". *International Journal of Heat and Mass Transfer*, Vol. 392, 134696.
3. **Bar-Kohany, T.**, and Jain, A. (2025) "[Heat transfer modeling during the readout process of a Thermoluminescent Dosimeter \(TLD\)](#)", *Nuclear Engineering and Design*, Vol. 433, 113817.
4. Shalel, A., Katoshevski, D., **Bar-Kohany, T.**, (2025) "[Methodology for Analyzing powder-based fire extinguishing and its optimization](#)", *Fire*, Vol. 8(1), 22.
5. **Bar-Kohany, T.**, and Jain, A. (2024) "[Dissipation of boundary effects in multi-layer heat conduction problems](#)", *International Journal of Heat Transfer*, Vol. 223, 125207.
6. Nahon, L., **Bar-Kohany, T.**, Aharon, Y., Rabinovich, E., Biton, A., and Gilad, E. (2024) "[The effect of the radial heat loss, upper reservoir temperature, and corrugated wall channel on the flooding critical heat flux](#)", *Progress in Nuclear Energy*, Vol. 168, 105038.
7. Lankry, A., Koyfman, A., Haustein, H., Nir, Y., Ziskind, G., and **Bar-Kohany, T.**, (2024). "[Onset of heterogeneous nucleation in pool boiling of HFE-7100 following rapid heating on a microscale heater](#)", *Experimental Thermal and Fluid Science*, Vol. 152, 111125.
8. Arogeti, M., Sher, E., and **Bar-Kohany, T.** (2024). [An experimental study of the events that follow a drop hitting a small surface](#), *Experimental Thermal and Fluid Science*, Vol. 150, 111067.
9. Avni, O., Dagan, Y., **Bar-Kohany, T.**, and Sher, E. (2023) [Bubble dynamics under negative pressures: A missing link?](#) *Thermal Science and Engineering Progress*, Vol. 46, 102162.
10. **Bar-Kohany, T.**, Arogeti, M., Malka, A., and Sher, E. (2023). [Advances in liquid atomization via flash boiling – A global overview](#). *Energies*, Vol. 16(19), 6763.

11. **Bar-Kohany, T.**, Antonov, D. V., Strizhak, P. A., & Sazhin, S. S. (2023). [Nucleation and bubble growth during puffing and micro-explosions in composite droplets](#). *Fuel*, Vol. 340, 126991.
12. Skripov, P. V., **Bar-Kohany, T.**, Antonov, D. V., Strizhak, P. A., & Sazhin, S. S. (2023). [Approximations for the nucleation temperature of water](#). *International Journal of Heat and Mass Transfer*, Vol. 207, 123970
13. Avni, O., **Bar-Kohany, T.**, and Sher, E. (2022). [Flash boiling atomization triggered and driven by intensive radiation](#). *Thermal Science and Engineering Progress*, Vol. 32, 101334.
14. **Bar-Kohany, T.**, and Sher, E. (2021). [Flash boiling atomization under negative pressure conditions](#). *Atomization and Sprays*, Vol. 31(8), pp.1-8.
15. **Bar-Kohany, T.** (2021). "[Minimal heating rate for isobaric nucleation at the spinodal in liquids](#)." *International Journal of Heat and Mass Transfer*, Vol. 165, 120636.
16. Sazhin, S. S., **Bar-Kohany, T.**, Nissar, Z., Antonov, D., Strizhak, P. A., & Rybdylova, O. D. (2020). "[A new approach to modelling micro-explosions in composite droplets](#)." *International Journal of Heat and Mass Transfer*, Vol. 161, 120238.
17. Arogeti, M., Sher, E., and **Bar-Kohany, T.**, (2019), "[Drop impact on small targets with different target-to-drop diameters ratio](#)", *Chemical Engineering Science*, Vol. 193, pp.89-101.
18. **Bar-Kohany, T.** and Amsalem, Y., (2018), "[Nucleation temperature under various heating rates](#)", *International Journal of Heat and Mass Transfer*, Vol. 126, pp. 411-415.
19. Dagan, Y. and **Bar-Kohany, T.**, (2018), "[Flame propagation through three-phase methane-hydrate particles](#)", *Combustion and Flames*, Vol. 193, pp.25-35.
20. Arogeti, M., Sher, E. and **Bar-Kohany, T.**, (2017), "[A single drop impact on a flat, dry surface – A unified correlation](#)", *Atomization and Sprays*, Vol.27(9), pp.759-770.
21. Eluk, T., Levy, A., Korytnyi, E. and **Bar-Kohany, T.**, (2017) "[Transition Mechanism between combustion regions in swirling flow in downer reactors](#)", *Energy and Fuels*, Vol.31(2),pp.1927-1934.
22. **Bar-Kohany, T.** and Levy, M., (2016), "[State of the art review of flash boiling atomization](#)", *Atomization and Sprays*, Vol. 26(12), pp. 1259-1305.

23. **Bar-Kohany, T.** and Sirignano, W.A., (2016), "[Transient combustion of a methane-hydrate sphere](#)", *Combustion and Flames*, Vol.163, pp.284-300.
24. Cohen, I., **Bar-Kohany, T.**, German, U. and Ziskind, G., (2014), "[Optimization of the temperature profiles due to a nitrogen jet impinging on a TLD detector](#)", *Radiation and Measurements*, Vol.70, pp. 48-51.
25. Sher, E., Sher, I. and **Bar-Kohany, T.**, (2013), "[Another view of the upper and intermediate explosion limits of a H₂-O₂ system.](#)", *Elsevier, International Journal of Hydrogen Energy* 38, pp.14912-14914, 2013
26. **Bar-Kohany, T.** and Dahan, K., (2012), "[Evaluation of the one-step hydrogen-oxygen global reaction rate in a non-premixed mixture to predict auto-ignition limits](#)", *International Journal of Hydrogen Energy*, Vol.37(19), pp.14669-14675.
27. **Bar-Kohany, T.** and Stern, A., (2011), "[Prediction of Container Hermeticity for Isostatic Pressuring Process](#)", *Research Bulletin of the Australian Institute of High Energetic Materials*, Vol.1, pp.10-27.
28. **Bar-Kohany, T.**, Rozenfeld, R., Weinstein, M., Abraham, A., German, U., Alfassi, Z.B. and Ziskind, G., (2011), "[Thermal Behavior of a LiF Crystal Mounted in a TLD Card and Heated by Jet Impingement](#)", *Radiation and Measurements*, Vol.46(12), pp.1432-1435.
29. Sher, E. ,**Bar-Kohany, T.** and Rashkovan, A., (2008), "[Flash Boiling Atomization](#)", *Invited Review, Progress in Energy and Combustion Science (PECS)*, Vol.34(4), pp.417.
30. **Bar-Kohany, T.**, I. Sher and E. Sher, (2007), "[Choked Flow of a Two-Phase Mixture Through and Aperture: a Theoretical Approach](#)", *Atomization and Sprays*, Vol.17(5), pp.431-449.
31. **Bar-Kohany, T.** and Stern, A., (2007), "[Lifetime Estimation of MOEMS Devices](#)", *Journal of Electronic Packaging*, Vol.129, pp.144-148.
32. **Bar-Kohany, T.** and Sher, E. (2004), "[Effervescent Atomization Under Sub-sonic and Choked Conditions – a Theoretical Approach](#)", *Chem. Eng. Sci.*, Vol. 59, pp.5987-5995.
33. **Bar-Kohany, T.** and Sher, E. (2004), "[Subsonic Atomization Effervescent: a Theoretical Approach](#)", *Atomization and Sprays*, Vol. 14(6), pp.495-509.
34. **Bar-Kohany, T.** and Sher, E. (2002), "[Optimization of Variable Valve Timing for Maximizing Performance of an Unthrottled SI Engine – a Theoretical Study](#)", *Energy*, Vol. 27, pp.757-775.
35. **Bar-Kohany, T.** and Sher, E. (1999), "[Using the 2nd Law of Thermodynamics to Optimize Variable Valve Timing for Maximizing Torque in a Throttled SI Engine](#)", *SAE Technical Papers*, 1999-01-0328.

Accepted for publication

Propriety papers

36. Over 50 NRCN, propriety papers

International Conferences, selected papers

1. **Bar-Kohany, T.**, Biton, D., Koyfman, A., Vaknin, M., and Ziskind, G. A Novel Non-Intrusive Method for Detecting CHF during Rapid Transients: Application to Spray and Immersion Cooling, 34th *ILASS*, Sept. 6-9, Lisbon, Portugal, 2026.
2. Magen, O., Kozak, Y., Banerjee, N., Marengo, M., and **Bar-Kohany, T.** A Theoretical and Numerical Framework to Predict the Onset of Nucleate Boiling During Rapid Depressurization, 34th *ILASS*, Sept. 6-9, Lisbon, Portugal, 2026.
3. **Bar-Kohany, T.**, Under Pressure: Navigating Thermodynamics and the Realization of Tensioned Liquids, *DIPSI Workshop on Droplet Impact Phenomena & Spray Investigation*, June 19th, Bergamo, Italy, 2026
4. Magen, O., Kozak, Y., Di Luccio, L., Marengo, M., and **Bar-Kohany, T.** An analytical thermodynamic model for flash boiling atomization under rapid depressurization, validated against cryogenic and refrigerant experiments, 33th *ILASS*, Aug.31-Sept. 4, Lund, Sweden, 2025.
5. Amsalem, Y., Lankry, A., Vaknin, M., Arogeti, M., and **Bar-Kohany, T.** An experimental setup for impingement of volatile liquids and liquid gallium, 33th *ILASS*, Aug.31-Sept. 4, Lund, Sweden, 2025.
6. Magen, O., Kozak, Y., Di Luccio, L., Marengo, M., and **Bar-Kohany, T.** A predictive Framework for the Onset of Flash Boiling Under Positive and Negative Pressures, *DIPSI Workshop on Droplet Impact Phenomena & Spray Investigation*, June 12th, Bergamo, Italy, 2025.
7. Amsalem, Y., Lankry, A., Vaknin, M., Arogeti, M., and **Bar-Kohany, T.** Impingement of liquid droplets – A unique experimental setup, *DIPSI Workshop on Droplet Impact Phenomena & Spray Investigation*, June 12th, Bergamo, Italy, 2025.
8. Sher, E., and **Bar-Kohany, T.** Overview of the development and advances in Liquid atomization by flash boiling. 32th *ILASS*, September 4-7, Napoli, Italy, 2023.
9. Sazhin, S.S. Antonov, D.V. Strizhak, P.A. Castanet, G. Shchepakina, E.A. Sobolev, V.A. **Bar-Kohany, T.**, Heating and evaporation of composite multicomponent droplets: recent experimental and modelling results, 17th International Heat Transfer Conference (*IHTC*), Cape-Town, South Africa, August 14-18, 2023.
10. **Bar-Kohany, T.**, Anotonov, D., Strizhak, P.A., Sazhin, S., “Nucleation in a rapidly heating of composite droplets: Experimental observations and modelling”, 11th International Conference of Boiling and Condensation Heat Transfer (*ICBCHT*), Edinburgh, UK, May 15-17, 2023.
11. **Bar-Kohany, T.**, “Slow and rapid heterogeneous nucleation of HFE-7100 at one atmosphere”, 8th *ASTFE* conference, UMD, College Park, March 23-26, 2023.
12. Sazhin, S., Anotonov, D., Strizhak, P.A., Castanet, G., Schepakina, E.A., Sobolev, V.A., and **Bar-Kohany, T.**, “Puffing and micro-explosions in composite droplets: recent modelling and results”, *DIPSI Workshop on Droplet Impact Phenomena & Spray Investigation*, July 1st, Bergamo, Italy, 2022.
13. Avni, O., **Bar-Kohany, T.**, and Sher, E., “On the prospect of rapid heating by intensive laser radiation for flash boiling atomization”, 31th *ILASS*, September 6-8. Online, 2022.
14. **Bar-Kohany, T.**, Anotonov, D., Strizhak, P.A., and Sazhin, S., “Nucleation and bubble growth in a rapidly heated composite drop”, 31th *ILASS*, September 6-8, Online, 2022.
15. Sazhin, S., **Bar-Kohany, T.**, Anotonov, D., and Strizhak, P.A., “Micro-explosions: Simple models of complex phenomena”, 10th *ISFEH, International Seminar on Fire and Explosion Hazards*, Oslo, Norway, 22-27 May, 2022.
16. Lankry, A., Koyfman, A., Korotkin, Y., Reuven, R., Haustein, H., Ziskind, G., and **Bar-Kohany, T.**, “Onset of heterogeneous nucleation of HFE-7100 on a rapidly heated micro-heater at one atmosphere”, 31th *ICLASS*, 30 Aug.-2 Sept. Online, 2021.

17. Arogeti, M., Sher, E., Asis, E., and **Bar-Kohany, T.**, "Experimental study of the sequel events after drop impacting a small target", 30th ILASS, September 6-8 Tel-Aviv, Israel, 2020.
18. Shalel, A., **Bar-Kohany, T.**, Katoshevski, D., "Multi-scale powder dispersal solution using the General Dynamic Equation", 29th ILASS, September 2-4, Paris, France, 2019.
19. **Bar-Kohany, T.**, Amsalem, Y., "Nucleation inception temperature in boiling due to rapid heating – A universal correlation", 29th ILASS, September 2-4, Paris, France, 2019.
20. Nahon, L., **Bar-Kohany, T.**, Rabinovich, E., and Aharon, Y., "Critical heat flux in vertical annulus under zero flow conditions with radial heat losses", NURETH-18, August 18-23, USA, 2019.
21. **Bar-Kohany, T.**, "Prediction of Nucleation Temperature under Moderate to High Heating Rates", Gordon Research Conference (GRC), February 3-8, Italy, 2019.
22. ¹**Bar-Kohany, T.**, Haustein, H.D., and Elias, E., "Lifetime of a tensioned liquid following a depressurization impulse", 14th ICLASS, July 22-26, Chicago, USA, 2018.
23. ¹**Bar-Kohany, T.**, and Dagan, Y., "Evaporation and combustion of spray consisting of three-phase methane-hydrate particles", 14th ICLASS, July 22-26, Chicago, USA, 2018.
24. ²**Bar-Kohany, T.**, "Combustion of three-phase methane-hydrate spheres", *Joint aerospace and fluid dynamics reviews seminars*, University of Maryland, College Park, 2018 – **Invited Lecture**
25. ¹**Bar-Kohany, T.**, Sirignano, W.A., "Combustion of a methane-hydrate particle", 27th ILASS, September, 4-7, 2016, Brighton, UK.
26. **Bar-Kohany, T.**, "Mechanism of spray formation by bi-component liquid flashing", TAU-UMD joint workshop on single and multiphase heat transfer, July, 25-29, Tel-Aviv, 2016.
27. Eluk, T., Korytnyi, E., Levy, A. and **Bar-Kohany, T.**, "Transition between robust equilibrium design points for multiphase combustion in downers", 9th International conference on multiphase flow, May, 22-27, 2016, Firenze, Italy.
28. **Bar-Kohany, T.** and Dahan, K., "Hydrogen Release, Dispersion and Auto-Ignition In Enclosures", 7th Mediterranean Combustion Science (MCS), September 11-15, 2011, Sardinia, Italy.
29. **Bar-Kohany, T.** and E. Sher, "Critical Discharge from Flash Boiling Atomizer", 10th Joint European Thermodynamic Conference (JETC), June 22-24, 2009, Copenhagen, Denmark.
30. **Bar-Kohany, T.** and E. Sher, "Spray Formation by Bi-Component Liquid Flashing Under Sub- and Choked Conditions", International Conference of Heat and Mass Transfer (ICHMT), June 5-10, 2005, Antalya, Turkey.
31. A. Stern, M. Kemelman, **T. Bar-Kohany**, and K. Feldman, "Hermetic Encapsulation of MOEMS by Fluxless Soldering", *Welding and Joining*, January 25-28, 2005, Tel-Aviv, Israel
32. **Bar-Kohany, T.** and E. Sher, "Spray Formation by Bi-Component Liquid Flashing Under Sub- and Choked Conditions: a Theoretical Approach", ILASS, September 6-8, 2004, Nottingham, UK.
33. **Bar-Kohany, T.** and E. Sher, "Spray Formation by Bi-Component Liquid Flashing: a Theoretical Approach", *International Conference on Liquid Atomization and Spray Systems (ICLASS)*, July 13-17, 2003, Sorrento, Italy, oral presentation.
34. **Bar-Kohany, T.** and E. Sher, "Spray Formation by Bi-Component Liquid Flashing: a Theoretical Approach", *European Conference of the Institution of Liquid Atomization and Spray Systems (ILASS)*, September 9-11, 2002, Zaragoza, Spain.
35. **Bar-Kohany, T.** and E. Sher, "Using the 2nd Law of Thermodynamics to Optimize Variable Valve Timing for Maximizing Torque in a Throttled SI Engine", *SAE International*, March 1-4, 1999, Detroit, Canada.

¹ Session Chair & Lecturer

² Invited lecture

National Conferences

1. Magen, O., Kozak, Y., Di Lucchio, L., Marengo, M., **Bar Kohany, T.**, "A Unified thermodynamic model for predicting flash-boiling inception", The Annual Israeli SAF Conference, September, 2025, Technion, Haifa, Israel
2. Magen, O., Kozak, Y., Di Lucchio, L., Marengo, M., **Bar Kohany, T.**, "Onset of Flash Boiling Due to Rapid Depressurization Across Positive and Negative Ranges – A Theoretical Model", 37th Annual Symposium of the Israeli Section of the Combustion institute (ISCI-37), January, 2025, Tel-Aviv, Israel
3. **Bar Kohany, T.**, "Study of the effect of rapid boiling on the analysis of nuclear accidents", Round table – Physical and Engineering studies, June, 2024, Tel-Aviv University, Israel.
4. Nahon, L. **Bar Kohany, T.**, Aharon, Y., Rabinovich, E., Biton, A., Gilad, A., "Experimental Study of flooding critical heat flux in vertical annular channels with no flow", 30th Conference of Nuclear societies in Israel, 7-9, June, 2022 Herzeliya, Israel.
5. Avni, O., **Bar-Kohany, T.**, and Sher, E., 35th Annual Symposium of the Israeli Section of the Combustion institute (ISCI-35), December, 2021, Tel-Aviv, Israel.
6. Shalel, A., **Bar-Kohany, T.**, Katoshevski, D., "Fire Powder extinguishing analysis using the Extended General Dynamic Equation", 33th Annual Symposium of the Israeli Section of the Combustion institute (ISCI-33), December, 2019, Tel-Aviv, Israel.
7. **Bar-Kohany, T.**, "Onset of Nucleation Boiling due to Rapid Heating – How does it relate to Combustion", 33th Annual Symposium of the Israeli Section of the Combustion institute (ISCI-33), December, 2019, Tel-Aviv, Israel.
8. **Bar-Kohany, T.**, and Dagan, Y., "Combustion of spray consisting of three-phase particles", 32th Annual Symposium of the Israeli Section of the Combustion institute (ISCI-32), December, 2018, Tel-Aviv, Israel.
9. Amsalem, Y., and **Bar-Kohany, T.**, "Combustion of spray consisting of three-phase particles", 35th Israeli Conference of Mechanical Engineering (ICME-35), 9-10 October, 2018, Ben-Gurion University, Israel.
10. Shalel, A., **Bar-Kohany, T.**, and Katoshevski, D., (2017), "Optimization of extinguishing powder dispersal", The 27th Annual meeting of the Israeli Association of Aerosol Research (IAAR-27), March 13, Technion, Israel.
11. **Bar-Kohany, T.**, "Depressurization of liquid to low or negative pressure; What is it good for?", 34th Israeli Conference of Mechanical Engineering (ICME-34), 21-22 November, 2016, Technion, Israel.³
12. Eluk, T., Levy, A., Kortnyi, E., and **Bar-Kohany, T.**, "Controlling robust design points for gas-solid combustion in co-current reactors", 34th Israeli Conference of Mechanical Engineering (ICME-34), 21-22 November, 2016, Technion, Israel.³
13. Eluk, T., Levy, A., Kortnyi, E., and **Bar-Kohany, T.**, "Transition between robust equilibrium design points for gas-solid combustion on co-current reactors", 40th Symposium on computational mechanics (ISCM-40), April, 7, 2016, Tel-Aviv University, Israel
14. **Bar-Kohany, T.**, and Aharon, Y., "Characteristic times and fuel response to a sudden increase in core reactivity", 28th Conference of Nuclear societies in Israel, 12-14, October, 2016 Dead-Sea, Israel.
15. Dahan, K., and **Bar-Kohany, T.**, "", 33th Israeli Conference of Mechanical Engineering (ICME-33), 2015, Technion, Israel.
16. Cohen, I., German, U., Ziskind, G., and **Bar-Kohany, T.**, "Parametric analysis of temperature profiles due to a nitrogen jet impinging on a TL Detector", 33th Israeli Conference of Mechanical Engineering (ICME-33), 2015, Technion, Israel.

³ Session chair & Lecturer

17. **Bar-Kohany, T.**, and Sirignano, W.A., "Combustion of a methane-hydrate particle", 28th Annual Symposium of the Israeli Section of the Combustion Institute, December, 2014, Tel-Aviv, Israel
18. Cohen, I., German, U., Ziskind, G., and **Bar-Kohany, T.**, "Aanalysis of temperature profiles due to a nitrogen jet impinging on a TLD", 26th Annual Symposium of the Israeli Section of the Combustion Institute, 21-23 February, 2012, Tel-Aviv, Israel
19. Rozenfeld, R., **Bar-Kohany, T.**, Weinstein, M., Abraham, A., German, U., Alfassi, Z.B. and Ziskind, G., "Study of thermal behavior of a TLD detector due to heating by Jet impingement", *The 26th Conference of the Nuclear Societies in Israel*, February, 2012, Dead-Sea, Israel
20. Rozenfeld, R., **Bar-Kohany, T.**, Weinstein, M., Abraham, A., German, U., Alfassi, Z.B. and Ziskind, G., "Thermal Behavior of a TLD Detector Due to Heating by Jet Impingement", *The 25th Conference of the Nuclear Societies in Israel*, February, 2010, Dead-Sea, Israel
21. **Bar-Kohany, T.** and Sher, E., "Author look into the 3 Explosion Limits", *24th Annual Symposium of the Israel Section of the Combustion Institute*, December, 2009, Tel-Aviv, Israel
22. **Bar-Kohany T.**, Feldman, K. and A. Stern, "Optical Micro-electro-mechanical Systems – Wafer Level Encapsulation", *The 12th Israeli Materials Engineering Conference*, March 2006, Tel-Aviv, Israel.
23. **Bar-Kohany T.** and A. Stern, "Leak Testing of MOEMS – Theory and Practice", 30th Israel Conference on Mechanical Engineering, May 29-30, 2005, Tel-Aviv, Israel.
24. Sher, E., Rashkovan, A. and **Bar-Kohany, T.**, "Effervescent Spray", *21th Annual Symposium of the Israel Section of the Combustion Institute*, December, 2005, Technion, Haifa, Israel
25. **Bar-Kohany, T.** and E. Sher, "A Study of the Mechanism of Sprays Formation by Bi-component Liquid Flashing", *30th Israel Conference on Mechanical Engineering*, May 29-30, 2005, Tel-Aviv, Israel
26. **Bar-Kohany, T.** and E. Sher, "Spray Formation by Bi-Component Liquid Flashing: a Theoretical Approach", *20th Israeli Conference on Combustion*, Beer-Sheva, Israel, December 9, 2004.
27. **Bar-Kohany T.** and A. Stern, "Hermeticity Tests of Sealed Cavities", *ASNT-Israel*, July 14, 2004.
28. **Bar-Kohany T.** and A. Stern, "Non-Destructive Leak Tests in Hermetic Sealed Cavities – Practical and Theoretical Considerations", *ISME- Israel National Conference on Welding*, March, 2004.
29. **Bar-Kohany, T.** and E. Sher, "A Study of Spray Formation by Bi-component Liquid Flashing", *29th Israel Conference on Mechanical Engineering*, May 12-13, 2003, Haifa, Israel
30. **Bar-Kohany, T.** and E. Sher, "Variable Valve Timing for Gasoline Engines", *28th Israel Conference on Mechanical Engineering*, June 14-15, 2000, Beer-Sheva, Israel

Additional Professional Activities

Positions in academic administration

1. 2025-present, Guest Editor, International Journal of Heat and Mass Transfer
2. 2022-present, ⁴ILASS-Europe Committee member
3. 2022, Co-Chair of the 30th ILASS-Europe conference.
4. 2020-2021, Guest Editor, Atomization and Sprays.
5. 2020, Co-Chair of the 29th ILASS-Europe conference.
6. 2018-present: Scientific committee member of the ILASS & ICLASS conferences.

⁴ ILASS - Institution of Liquid and Atomization Spraying Systems

ICLASS – International Conference on Liquid Atomization and Spraying Systems

7. 2019-2023: Scientific committee member of the PAIC⁵.
8. 2012 – present: PhD & MSc Academic Judging committees, BGU

Significant professional consulting

2023 - 2026, WatAirGy, Spray & Thermodynamic analysis.

2014, Therm-Sphera, Thermodynamic process optimization.

Ad-hoc reviewer for journals (by alphabetic order)

Atomization and Sprays (AAS), Chemical Engineering Science (ChES), Chemical Review Engineering (CRE), Energies, Experiments in Fluids (EF), Experimental Thermal and Fluids Science (ETFS), Fuel, International Journal of Hydrogen Energy (IJHE), International Journal of Heat and Mass Transfer (IJHMT), International Journal of Multiphase Flow(IJMF), Journal of Aerosol Science (JAS), Journal of Fluid Mechanics (JFM), Journal of Heat Transfer (JHT), Physics of Fluids (PF), Scientific Reports, etc.

Research grants

2025-2028, **Bar-Kohany^{PI}, T.**, Ullman A.^{PI}, Department of Energy (DoE-IL), Title: “Boiling following rapid heating in pool and flow boiling - various orientations”, annual: 248 kNIS (66 k€), total: 744 kNIS (198 k€).

2024-2027, **Bar-Kohany^{PI}, T.**, Marengo M.^{PI}, E., Ministry of Innovation, Science and Technology (MOST), Title: “Rapid phase-change of hydrogen and hydrogen carrier fuels (HFCs)”, annual: 64 k€, total: 192 k€.

2017-2021, **Bar-Kohany^{PI}, T.**Haustein^{PI}, H. and Elias^{CI}, E., Council of higher education (VATAT), Title: “Non-equilibrium aspects of boiling related to safety scenarios in nuclear reactors”, annual: 100 k\$, total: 400 k\$.

2013-2017, **Bar-Kohany^{PI}, T.** and Levy^{PI}, A., Council of higher education (VATAT), Title: “Co-current gas-solid reactive flow in a downer flame reactor”, annual: 57 k\$, total: 228 k\$.

⁵ PAIC – Practical Aspects of Chemical Engineering